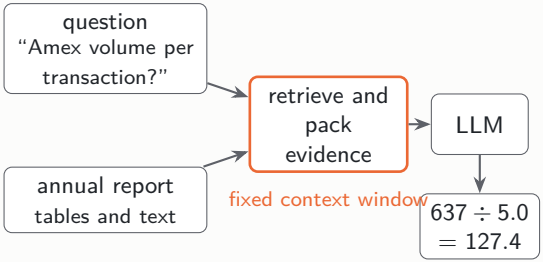
Dependency-Preserving Evidence Packing for Numerical Question Answering over Financial Reports

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Six table–text QA datasets | PySpark | Gemini 3.5 Flash-Lite

The task: numerical question answering over reports



- An NLP task: table–text hybrid question answering, with arithmetic.
- The model reads only what fits in its **context window**; a HybridQA document is 8,928 tokens at the median, the budget 1,500.
- So the system must choose which facts to pack, and **how to write them**.

The problem: a number means nothing without its labels

company	paym. vol. (bn)	total vol. (bn)	transactions (bn)	cards (mn)
mastercard	1697	2276	27.0	916
american express	637	647	5.0	86
discover	102	119	1.6	57

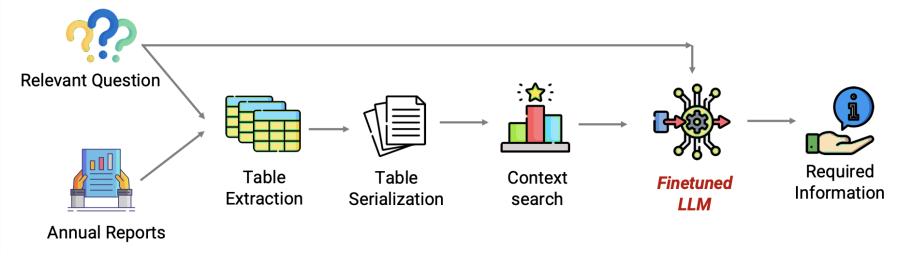
FinQA development set, V/2008/page_17 (Visa annual report).

The retriever's top two cells are **637** and **647**. Only the column headers separate them.

- **Drop labels**: cheap, but numbers become ambiguous.
- **Repeat labels per cell**: readable, but a cell costs **4–8**× more tokens (15 vs 3.6 on FinQA, 24.7 vs 3.0 on MultiHiertt); needed numbers fall out of the window.

We need packing that keeps every number's labels and still fits the context window.

Prior approach: serialize tables with labels in every sentence



Pipeline of Arun et al., *Numerical Reasoning for Financial Reports*, arXiv:2312.14870 (2023), their Fig. 4.

- Tables become sentences: "the revenue of 2019 is 100".
- FAISS keeps one chunk; a fine-tuned T5 / Llama-2 answers.
- Labels survive because they are **repeated with every number**: the cost our method removes.
- Their lesson we keep: let code do the arithmetic.

Our contribution

Keep every number's labels, but **write each label once** and let the numbers point to it.
The same budget then holds more numbers, and every one stays readable.

1. **Dependency-preserving packing**, with its exact token trade-off.
2. **A PySpark data layer**: one verified graph of 1.36 M facts and 1.88 M label links.
3. **A coverage audit** on all 9,268 dev questions, no model.
4. **A paired answer test** with Gemini 3.5 Flash-Lite and grounded answers.

Objectives

	Objective	Evidence
O1	One pipeline over the four assigned datasets plus TAT-HQA and MultiHiertt, with official scorers	gold answers through each scorer
O2	Dependency packing against label-repeating and length-matched packing at a fixed budget	coverage audit, 9,268 questions
O3	A verified PySpark preparation of all six datasets	equality with a Python reference
O4	A paired answer test with one model	Block A, 50 questions per dataset
O5	Answers grounded in the evidence	Block B, executor vs direct

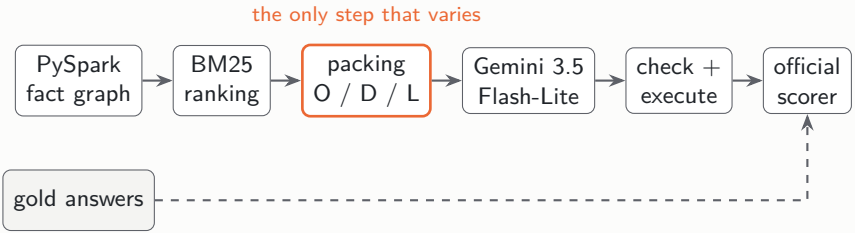
Datasets

	Dataset	Dev questions	Challenge	Median document (tokens) labels once	labels per cell	Budget
Assigned	FinQA	883	arithmetic programs	1,041	1,112	800
	ConvFinQA	1,490 turns	conversation history	979	1,062	800
	TAT-QA	1,668	spans, scale	396	592	400
	HybridQA	3,466	cell-to-passage links	8,928	11,142	1,500
Added	TAT-HQA	824	hypothetical changes	396	592	400
	MultiHiertt	1,044	hierarchical tables	3,693	6,435	1,500

Official GitHub repositories at pinned commits: czyssrs/FinQA, czyssrs/ConvFinQA, NExTplusplus/TAT-QA, wenhuchen/HybridQA, NExTplusplus/TAT-HQA, psunlpgroup/MultiHiertt.

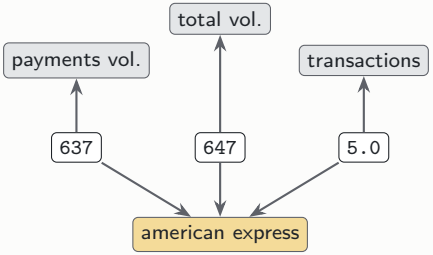
- Related datasets share documents, so results are never pooled.
- TAT-QA's median document fits its 400-token budget only when labels are written once.

Pipeline



- Every dataset becomes *facts*: text plus the IDs of the facts needed to read it.
- BM25 ranks cells and paragraphs; packing fills a fixed token budget.
- One model call per question; the model copies numbers into a plan.
- Gold answers meet predictions only at scoring: 0 of 443 prompts contained one.

Dependency graph and packing



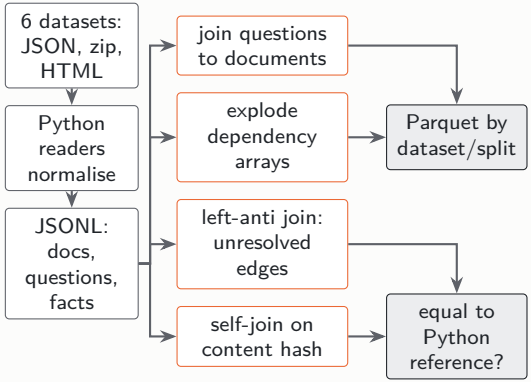
Each number points to the labels it needs; the row label is one shared node.

- **Ordinary (O)**: labels inline in every cell.
- **Dependency (D)**: a fact enters with its label closure; packed labels are free.
- **Length-matched (L)**: ordinary facts plus neighbours, to D's length.

Amex cells packed	1	2	3	4
O tokens	15	29	45	57
D tokens	18	29	42	50

A fixed cost for structure, then cheaper cells: 15.1 vs 3.6 tokens per FinQA cell, 24.7 vs 3.0 in MultiHiertt.

Big Data implementation with PySpark



- Spark 4.3.0, Spark SQL DataFrames, local mode, 8 shuffle partitions.
- Output: **1,360,644 facts**, **1,884,748 edges**, 0 unresolved; 4,395 deduplicated documents.
- Self-join finds **491 identical document pairs**: why datasets are never pooled.
- Parquet 0.18 GB vs 0.95 GB JSON lines (5.1×).
- Full data: Spark 37.0 s at 4 cores, Python 15.8 s. At 1.4 M in-memory rows Spark gives a verified columnar store, not speed.

Grounded answers

- The model returns a JSON plan whose arguments are numbers **copied** from the evidence: `divide("637", "5.0")`.
- Code finds each number in the evidence; a missing number rejects the plan.
- A bounded decimal interpreter computes the answer.

- Blocks invented numbers: 640 is in no cell.
- Cannot block a wrong real number: 647 passes and gives 129.4.
- That choice is only safe when the labels are in the evidence.

Experimental setup

Three measurements

- **Coverage audit**, no model: does every gold operand survive packing? 9,268 questions $\times$ 5 budgets $\times$ 3 methods.
- **Block A**: O vs D vs L answers, 50 questions per dataset.
- **Block B**: executor vs direct answering on D evidence, 40 per dataset.

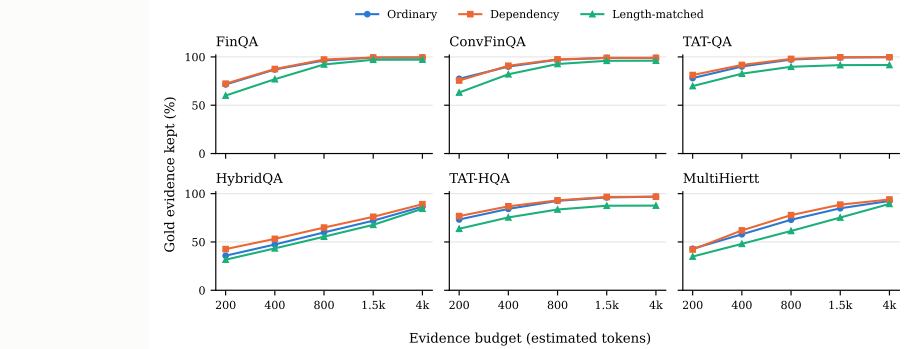
Budgets set from document sizes: 400, 800 or 1,500 tokens.

Model configuration

	Block A	Block B
Model	Gemini 3.5 Flash-Lite	Gemini 3.1 Flash-Lite
Version	3.5-flash-lite-07-2026	3.1-flash-lite-05-2026
Training	none, as served	none, as served
Output	schema-bound JSON	schema-bound JSON
Max output / thinking	2,048 / low	2,048 / low
Responses	885	472
Median tokens in / out	1,403 / 129	1,214 / 109

Paired bootstrap over documents, 2,000 resamples; failures score zero.

Result 1: more of the needed evidence survives



D above O on all six datasets at the chosen budgets. Kept only by D vs only by O: HybridQA **146 vs 7**, MultiHiertt **42 vs 3**, TAT-QA 41 vs 15, TAT-HQA 33 vs 12, FinQA 9 vs 1, ConvFinQA 6 vs 1. The gap closes as budgets grow; at 200 tokens it reverses on ConvFinQA and MultiHiertt.

Result 2: answers follow the evidence

Block A, Gemini 3.5 Flash-Lite (%)

Dataset	n	O	D	L	D−O	95% CI
FinQA	50	62.0	64.0	66.0	+2.0	[+0.0, +6.2]
ConvFinQA	45	73.3	73.3	68.9	+0.0	[−6.0, +6.4]
TAT-QA	50	60.2	71.5	69.4	+11.3	[+4.1, +19.9]
HybridQA	50	54.9	60.1	49.8	+5.3	[−2.4, +13.0]
TAT-HQA	50	56.7	60.7	52.7	+4.0	[−6.4, +14.5]
MultiHiertt	50	47.0	56.8	41.0	+9.8	[+0.0, +20.0]

- D $\geq$ O on all six datasets; significant on **TAT-QA, +11.3** [+4.1, +19.9].
- Elsewhere within noise at $n = 50$.

A real case (TAT-QA, budget 400)

- O uses 398 tokens on 5 facts; "accrued compensation, 358" needs 17 more.
- The model: *"the evidence does not contain data for accrued compensation"*; abstains.
- D pays 33 tokens for labels once and 3 per cell: 358 fits; `divide(948, 358)` $= 2.65$, the gold answer.

Result 3: the executor

Block B, Gemini 3.1 Flash-Lite (%)

Dataset	n	Executor	Direct	Diff.	95% CI
FinQA	40	62.5	67.5	−5.0	[−17.5, +7.0]
ConvFinQA	36	75.0	75.0	+0.0	[−8.6, +7.3]
TAT-QA	40	65.2	58.7	+6.5	[−7.6, +20.7]
HybridQA	40	50.2	66.6	−16.4	[−30.6, −2.0]
TAT-HQA	40	60.0	56.7	+3.3	[−12.0, +19.3]
MultiHiertt	40	63.7	41.7	+22.1	[+2.5, +40.0]

- No consistent executor gain; direct answering leads on HybridQA (text answers, no arithmetic).
- MultiHiertt's +22.1 is a scoring artefact: −29.92 written for a gold −0.2992.
- One numeric rule for both sides: **+6.2** [−6.2, +20.0]. Claim withdrawn.
- The executor's value: every answer traces to evidence.

Conclusion

- **Problem**: large tables must fit a fixed context window; numbers without labels mean nothing.
- **Method**: write each label once and link every number to it.
- **Evidence**: more of what a question needs survives on all six datasets; the gain closes as budgets grow.
- **Answers**: at least as good with Gemini 3.5 Flash-Lite; significant on TAT-QA.
- **Big Data**: PySpark builds and verifies the 1.36 M-fact graph.

Future work

- Larger answer samples to narrow the intervals.
- Budget-aware closure: pay for a header only when several packed facts share it.
- Better header inference for nested tables.

Limits: small answer samples, inferred headers, estimated token counts, hosted models.